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1. Homogeneous Function

A function f of two independent variables x, y is said to be a homogeneous function of degree n if it can be put in either of the following two forms :

$$f(x, y) = x^n \phi\left(\frac{y}{x}\right), \text{ where } \phi \text{ denotes a function of } \frac{y}{x}$$

or $f(tx, ty) = t^n f(x, y)$, where t is any positive real number.

Similarly, a function f of three independent variables x, y, z is said to be a homogeneous function of degree n if it can be put in either of the following two forms :

$$f(x, y, z) = x^n \phi\left(\frac{y}{x}, \frac{z}{x}\right), \text{ where } \phi \text{ denotes a function of } \frac{y}{x} \text{ and } \frac{z}{x}$$

or $f(tx, ty, tz) = t^n f(x, y, z)$, where t is any positive real number.

The above definition can be extended to a function of any number of variables.

Example : The function

$$f(x, y) = \frac{x^4 + y^4}{x - y}$$

is a homogeneous function of degree 3, since

$$f(x, y) = \frac{x^4 + y^4}{x - y} = \frac{x^4 \left[1 + \left(\frac{y}{x}\right)^4\right]}{x \left[1 - \left(\frac{y}{x}\right)\right]} = x^3 \frac{\left[1 + \left(\frac{y}{x}\right)^4\right]}{\left[1 - \left(\frac{y}{x}\right)\right]} = x^3 \phi\left(\frac{y}{x}\right)$$

or alternatively,

$$f(tx, ty) = \frac{(tx)^4 + (ty)^4}{tx - ty} = \frac{t^4(x^4 + y^4)}{t(x - y)} = t^3 \frac{(x^4 + y^4)}{(x - y)} = t^3 f(x, y).$$

Similarly, the function

$$f(x, y, z) = \frac{x^3 + y^3 + z^3}{x + y + z}$$

is a homogeneous function of degree 2, since

$$f(x, y, z) = \frac{x^3 + y^3 + z^3}{x + y + z} = \frac{x^3 \left[1 + \left(\frac{y}{x}\right)^3 + \left(\frac{z}{x}\right)^3\right]}{x \left[1 + \left(\frac{y}{x}\right) + \left(\frac{z}{x}\right)\right]} = x^2 \frac{\left[1 + \left(\frac{y}{x}\right)^3 + \left(\frac{z}{x}\right)^3\right]}{\left[1 + \left(\frac{y}{x}\right) + \left(\frac{z}{x}\right)\right]} = x^2 \phi\left(\frac{y}{x}, \frac{z}{x}\right)$$

or alternatively,

$$f(tx, ty, tz) = \frac{(tx)^3 + (ty)^3 + (tz)^3}{tx + ty + tz} = \frac{t^3(x^3 + y^3 + z^3)}{t(x + y + z)} = t^2 \frac{(x^3 + y^3 + z^3)}{(x + y + z)} = t^2 f(x, y, z).$$

Note: A polynomial function is a homogeneous function of degree n if all of its terms are of the same degree n .

Proof : Let f be a polynomial function in two independent variables x, y , i.e.,

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \dots + a_{n-1} x y^{n-1} + a_n y^n.$$

$$\begin{aligned} \text{Then } f(x, y) &= x^n \left[a_0 + a_1 \frac{y}{x} + a_2 \frac{y^2}{x^2} + \dots + a_{n-1} \frac{y^{n-1}}{x^{n-1}} + a_n \frac{y^n}{x^n} \right] \\ &= x^n \left[a_0 + a_1 \left(\frac{y}{x} \right) + a_2 \left(\frac{y}{x} \right)^2 + \dots + a_{n-1} \left(\frac{y}{x} \right)^{n-1} + a_n \left(\frac{y}{x} \right)^n \right] \\ &= x^n \phi \left(\frac{y}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x}. \end{aligned}$$

Example : The function

$$f(x, y) = x^5 + 6x^4 y + 7x^3 y^2 + 2y^5$$

is a homogeneous function of degree 5, since

$$\begin{aligned} f(x, y) &= x^5 + 6x^4 y + 7x^3 y^2 + 2y^5 \\ &= x^5 \left[1 + 6 \left(\frac{y}{x} \right) + 7 \left(\frac{y}{x} \right)^2 + 2 \left(\frac{y}{x} \right)^5 \right] \\ &= x^5 \phi \left(\frac{y}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x}. \end{aligned}$$

Similarly, the function

$$f(x, y, z) = x^4 + 3x^2 y^2 + 4xyz^2 + 5yz^3 + 6y^4 + 7z^4$$

is a homogeneous function of degree 4, since

$$\begin{aligned} f(x, y, z) &= x^4 + 3x^2 y^2 + 4xyz^2 + 5yz^3 + 6y^4 + 7z^4 \\ &= x^4 \left[1 + 3 \left(\frac{y}{x} \right)^2 + 4 \left(\frac{y}{x} \right) \left(\frac{z}{x} \right)^2 + 5 \left(\frac{y}{x} \right) \left(\frac{z}{x} \right)^3 + 6 \left(\frac{y}{x} \right)^4 + 7 \left(\frac{z}{x} \right)^4 \right] \\ &= x^4 \phi \left(\frac{y}{x}, \frac{z}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x} \text{ and } \frac{z}{x}. \end{aligned}$$

2. Euler's Theorem on Homogeneous Function of Two Variables

Statement : If u be a homogeneous function of degree n in two independent variables x, y , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu.$$

Proof : Let

$$u = A_1 x^{\alpha_1} y^{\beta_1} + A_2 x^{\alpha_2} y^{\beta_2} + A_3 x^{\alpha_3} y^{\beta_3} + \dots + A_n x^{\alpha_n} y^{\beta_n} \quad \dots (1)$$

where $\alpha_1 + \beta_1 = \alpha_2 + \beta_2 = \alpha_3 + \beta_3 = \dots = \alpha_n + \beta_n = n$

Differentiating both sides of equation (1) partially w. r. t. x , we get

$$\frac{\partial u}{\partial x} = A_1 (\alpha_1 x^{\alpha_1-1}) y^{\beta_1} + A_2 (\alpha_2 x^{\alpha_2-1}) y^{\beta_2} + A_3 (\alpha_3 x^{\alpha_3-1}) y^{\beta_3} + \dots + A_n (\alpha_n x^{\alpha_n-1}) y^{\beta_n}$$

$$\text{This} \Rightarrow x \frac{\partial u}{\partial x} = A_1 \alpha_1 x^{\alpha_1} y^{\beta_1} + A_2 \alpha_2 x^{\alpha_2} y^{\beta_2} + A_3 \alpha_3 x^{\alpha_3} y^{\beta_3} + \dots + A_n \alpha_n x^{\alpha_n} y^{\beta_n} \quad \dots (2)$$

Now, differentiating both sides of equation (1) partially w. r. t. y , we get

$$\frac{\partial u}{\partial y} = A_1 x^{\alpha_1} (\beta_1 y^{\beta_1-1}) + A_2 x^{\alpha_2} (\beta_2 y^{\beta_2-1}) + A_3 x^{\alpha_3} (\beta_3 y^{\beta_3-1}) + \dots + A_n x^{\alpha_n} (\beta_n y^{\beta_n-1})$$

$$\text{This} \Rightarrow y \frac{\partial u}{\partial y} = A_1 \beta_1 x^{\alpha_1} y^{\beta_1} + A_2 \beta_2 x^{\alpha_2} y^{\beta_2} + A_3 \beta_3 x^{\alpha_3} y^{\beta_3} + \dots + A_n \beta_n x^{\alpha_n} y^{\beta_n} \quad \dots (3)$$

Adding equations (2) and (3), we get

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= (\alpha_1 + \beta_1) A_1 x^{\alpha_1} y^{\beta_1} + (\alpha_2 + \beta_2) A_2 x^{\alpha_2} y^{\beta_2} + (\alpha_3 + \beta_3) A_3 x^{\alpha_3} y^{\beta_3} \\ &\quad + \dots + (\alpha_n + \beta_n) A_n x^{\alpha_n} y^{\beta_n} \\ &= n A_1 x^{\alpha_1} y^{\beta_1} + n A_2 x^{\alpha_2} y^{\beta_2} + n A_3 x^{\alpha_3} y^{\beta_3} + \dots + n A_n x^{\alpha_n} y^{\beta_n} \\ &= n (\alpha_1 + \beta_1 = \alpha_2 + \beta_2 = \alpha_3 + \beta_3 = \dots = \alpha_n + \beta_n = n) \\ &= n (A_1 x^{\alpha_1} y^{\beta_1} + A_2 x^{\alpha_2} y^{\beta_2} + A_3 x^{\alpha_3} y^{\beta_3} + \dots + A_n x^{\alpha_n} y^{\beta_n}) \\ &= nu \text{ (using equation (1))} \end{aligned}$$

i.e.,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Corollary : If u be a homogeneous function of degree n in two independent variables x, y , then

$$(i) \quad x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = (n-1) \frac{\partial u}{\partial x}$$

$$(ii) \quad x \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} = (n-1) \frac{\partial u}{\partial y}$$

$$(iii) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u.$$

Proof : (i) Since u is a homogeneous function of degree n in two independent variables x, y , therefore, by Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad \dots\dots(1)$$

Differentiating both sides of equation (1) partially w. r. t. x , we get

$$x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}(1) + y \frac{\partial^2 u}{\partial x \partial y} = n \frac{\partial u}{\partial x}$$

$$\text{This } \Rightarrow x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = (n-1) \frac{\partial u}{\partial x} \quad \dots\dots(2)$$

Hence (i) is proved.

(ii) Differentiating both sides of equation (1) partially w. r. t. y , we get

$$x \frac{\partial^2 u}{\partial y \partial x} + y \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y}(1) = n \frac{\partial u}{\partial y}$$

$$\text{This } \Rightarrow x \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = n \frac{\partial u}{\partial y} \quad \left(\because \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \right)$$

$$\Rightarrow x \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} = (n-1) \frac{\partial u}{\partial y} \quad \dots\dots(3)$$

Hence (ii) is proved.

(iii) Multiplying equations (2) and (3) by x and y respectively and adding, we get

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= (n-1) \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \\ &= (n-1)(nu) \quad (\text{using equation (1)}) \end{aligned}$$

$$\text{i.e., } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u.$$

This proves (iii).

3. Euler's Theorem on Homogeneous Function of Three Variables

Statement : If u be a homogeneous function of degree n in three independent variables x, y, z , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu.$$

Proof : Let

$$u = A_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + A_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + A_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + A_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n} \quad \dots(1)$$

where $\alpha_1 + \beta_1 + \gamma_1 = \alpha_2 + \beta_2 + \gamma_2 = \alpha_3 + \beta_3 + \gamma_3 = \dots = \alpha_n + \beta_n + \gamma_n = n$

Differentiating both sides of equation (1) partially w. r. t. x , we get

$$\begin{aligned} \frac{\partial u}{\partial x} &= A_1 (\alpha_1 x^{\alpha_1-1}) y^{\beta_1} z^{\gamma_1} + A_2 (\alpha_2 x^{\alpha_2-1}) y^{\beta_2} z^{\gamma_2} + A_3 (\alpha_3 x^{\alpha_3-1}) y^{\beta_3} z^{\gamma_3} \\ &+ \dots + A_n (\alpha_n x^{\alpha_n-1}) y^{\beta_n} z^{\gamma_n} \end{aligned}$$

$$\text{This } \Rightarrow x \frac{\partial u}{\partial x} = A_1 \alpha_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + A_2 \alpha_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + A_3 \alpha_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + A_n \alpha_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n} \quad \dots(2)$$

Now, differentiating both sides of equation (1) partially w. r. t. y , we get

$$\begin{aligned} \frac{\partial u}{\partial y} &= A_1 x^{\alpha_1} (\beta_1 y^{\beta_1-1}) z^{\gamma_1} + A_2 x^{\alpha_2} (\beta_2 y^{\beta_2-1}) z^{\gamma_2} + A_3 x^{\alpha_3} (\beta_3 y^{\beta_3-1}) z^{\gamma_3} \\ &+ \dots + A_n x^{\alpha_n} (\beta_n y^{\beta_n-1}) z^{\gamma_n} \end{aligned}$$

$$\text{This } \Rightarrow y \frac{\partial u}{\partial y} = A_1 \beta_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + A_2 \beta_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + A_3 \beta_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + A_n \beta_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n} \quad \dots(3)$$

Similarly, differentiating both sides of equation (1) partially w. r. t. z , we get

$$\begin{aligned} \frac{\partial u}{\partial z} &= A_1 x^{\alpha_1} y^{\beta_1} (\gamma_1 z^{\gamma_1-1}) + A_2 x^{\alpha_2} y^{\beta_2} (\gamma_2 z^{\gamma_2-1}) + A_3 x^{\alpha_3} y^{\beta_3} (\gamma_3 z^{\gamma_3-1}) \\ &+ \dots + A_n x^{\alpha_n} y^{\beta_n} (\gamma_n z^{\gamma_n-1}) \end{aligned}$$

$$\text{This } \Rightarrow z \frac{\partial u}{\partial z} = A_1 \gamma_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + A_2 \gamma_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + A_3 \gamma_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + A_n \gamma_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n} \quad \dots(4)$$

Adding equations (2), (3) and (4), we get

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} &= (\alpha_1 + \beta_1 + \gamma_1) A_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + (\alpha_2 + \beta_2 + \gamma_2) A_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + \\ &(\alpha_3 + \beta_3 + \gamma_3) A_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + (\alpha_n + \beta_n + \gamma_n) A_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n} \\ &= n A_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + n A_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + n A_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + \\ &n A_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n} \\ &(\because \alpha_1 + \beta_1 + \gamma_1 = \alpha_2 + \beta_2 + \gamma_2 = \dots = \alpha_n + \beta_n + \gamma_n = n) \\ &= n (A_1 x^{\alpha_1} y^{\beta_1} z^{\gamma_1} + A_2 x^{\alpha_2} y^{\beta_2} z^{\gamma_2} + A_3 x^{\alpha_3} y^{\beta_3} z^{\gamma_3} + \dots + A_n x^{\alpha_n} y^{\beta_n} z^{\gamma_n}) \\ &= n u \text{ (using equation (1))} \end{aligned}$$

i.e.,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = n u$$

Example 1 : Verify Euler's Theorem when $u = \frac{x(x^3 - y^3)}{x^3 + y^3}$.

Solution : According to Euler's Theorem, if u be a homogeneous function of degree n in two independent variables x, y , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu.$$

Given that

$$u = \frac{x(x^3 - y^3)}{x^3 + y^3} \quad \dots\dots(1)$$

$$\text{i.e., } u = \frac{x^4 \left[1 - \left(\frac{y}{x} \right)^3 \right]}{x^3 \left[1 + \left(\frac{y}{x} \right)^3 \right]} = x \frac{\left[1 - \left(\frac{y}{x} \right)^3 \right]}{\left[1 + \left(\frac{y}{x} \right)^3 \right]} = x \phi \left(\frac{y}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x}.$$

This $\Rightarrow$ The given function u is a homogeneous function of degree 1 in two independent variables x, y . Therefore Euler's Theorem will be verified if we can prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u.$$

Taking logarithm of both sides of equation (1), we get

$$\log u = \log x + \log (x^3 - y^3) - \log (x^3 + y^3) \quad \dots\dots(2)$$

Now, differentiating both sides of equation (2) partially w. r. t. x , we get

$$\frac{1}{u} \frac{\partial u}{\partial x} = \frac{1}{x} + \frac{1}{x^3 - y^3} (3x^2) - \frac{1}{x^3 + y^3} (3x^2)$$

$$\text{This } \Rightarrow \frac{1}{u} \left(x \frac{\partial u}{\partial x} \right) = 1 + \frac{3x^3}{x^3 - y^3} - \frac{3x^3}{x^3 + y^3} \quad \dots\dots(3)$$

Similarly, differentiating both sides of equation (2) partially w. r. t. y , we get

$$\frac{1}{u} \frac{\partial u}{\partial y} = 0 + \frac{1}{x^3 - y^3} (-3y^2) - \frac{1}{x^3 + y^3} (3y^2)$$

$$\text{This } \Rightarrow \frac{1}{u} \left(y \frac{\partial u}{\partial y} \right) = -\frac{3y^3}{x^3 - y^3} - \frac{3y^3}{x^3 + y^3} \quad \dots\dots(4)$$

Adding equations (3) and (4), we get

$$\begin{aligned} \frac{1}{u} \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) &= 1 + \frac{3(x^3 - y^3)}{x^3 - y^3} - \frac{3(x^3 + y^3)}{x^3 + y^3} \\ &= 1 + 3 - 3 \\ &= 1 \end{aligned}$$

$$\text{This } \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$$

$\Rightarrow$ Euler's Theorem is verified for the given function.

Example 2 : Verify Euler's Theorem when $u = \frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{5}} + y^{\frac{1}{5}}}$.

Solution : According to Euler's Theorem, if u be a homogeneous function of degree n in two independent variables x, y , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu.$$

Given that

$$u = \frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{5}} + y^{\frac{1}{5}}} \quad \dots\dots(1)$$

$$\text{i.e., } u = \frac{x^{\frac{1}{4}} \left[1 + \left(\frac{y}{x} \right)^{\frac{1}{4}} \right]}{x^{\frac{1}{5}} \left[1 + \left(\frac{y}{x} \right)^{\frac{1}{5}} \right]} = x^{\frac{1}{20}} \frac{\left[1 + \left(\frac{y}{x} \right)^{\frac{1}{4}} \right]}{\left[1 + \left(\frac{y}{x} \right)^{\frac{1}{5}} \right]} = x^{\frac{1}{20}} \phi \left(\frac{y}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x}.$$

This $\Rightarrow$ The given function u is a homogeneous function of degree $\frac{1}{20}$ in two independent variables x, y . Therefore Euler's Theorem will be verified if we can prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{20} u.$$

Taking logarithm of both sides of equation (1), we get

$$\log u = \log \left(x^{\frac{1}{4}} + y^{\frac{1}{4}} \right) - \log \left(x^{\frac{1}{5}} + y^{\frac{1}{5}} \right) \quad \dots\dots(2)$$

Now, differentiating both sides of equation (2) partially w. r. t. x , we get

$$\frac{1}{u} \frac{\partial u}{\partial x} = \frac{1}{x^{\frac{1}{4}} + y^{\frac{1}{4}}} \left(\frac{1}{4} x^{-\frac{3}{4}} \right) - \frac{1}{x^{\frac{1}{5}} + y^{\frac{1}{5}}} \left(\frac{1}{5} x^{-\frac{4}{5}} \right)$$

$$\text{This } \Rightarrow \frac{1}{u} \left(x \frac{\partial u}{\partial x} \right) = \frac{x^{\frac{1}{4}}}{4 \left(x^{\frac{1}{4}} + y^{\frac{1}{4}} \right)} - \frac{x^{\frac{1}{5}}}{5 \left(x^{\frac{1}{5}} + y^{\frac{1}{5}} \right)} \quad \dots\dots(3)$$

Similarly, differentiating both sides of equation (2) partially w. r. t. y , we get

$$\frac{1}{u} \frac{\partial u}{\partial y} = \frac{1}{x^{\frac{1}{4}} + y^{\frac{1}{4}}} \left(\frac{1}{4} y^{-\frac{3}{4}} \right) - \frac{1}{x^{\frac{1}{5}} + y^{\frac{1}{5}}} \left(\frac{1}{5} y^{-\frac{4}{5}} \right)$$

$$\text{This } \Rightarrow \frac{1}{u} \left(y \frac{\partial u}{\partial y} \right) = \frac{y^{\frac{1}{4}}}{4 \left(x^{\frac{1}{4}} + y^{\frac{1}{4}} \right)} - \frac{y^{\frac{1}{5}}}{5 \left(x^{\frac{1}{5}} + y^{\frac{1}{5}} \right)} \quad \dots\dots(4)$$

Adding equations (3) and (4), we get

$$\begin{aligned}\frac{1}{u} \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) &= \frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{4 \left(x^{\frac{1}{4}} + y^{\frac{1}{4}} \right)} - \frac{x^{\frac{1}{5}} + y^{\frac{1}{5}}}{5 \left(x^{\frac{1}{5}} + y^{\frac{1}{5}} \right)} \\ &= \frac{1}{4} - \frac{1}{5} \\ &= \frac{1}{20}\end{aligned}$$

$$\text{This } \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{20} u$$

$\Rightarrow$ Euler's Theorem is verified for the given function.

Example 3 : If $u = \tan^{-1} \frac{x^2 + y^2}{x + y}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$.

Solution : Given that

$$u = \tan^{-1} \frac{x^2 + y^2}{x + y}.$$

$$\text{This } \Rightarrow \tan u = \frac{x^2 + y^2}{x + y} = \frac{x^2 \left[1 + \left(\frac{y}{x} \right)^2 \right]}{x \left[1 + \frac{y}{x} \right]} = x \frac{\left[1 + \left(\frac{y}{x} \right)^2 \right]}{\left[1 + \frac{y}{x} \right]} = x \phi \left(\frac{y}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x}.$$

$\Rightarrow \tan u$ is a homogeneous function of degree 1 in two independent variables x, y .

Let $v = \tan u$ (1)

Then v is a homogeneous function of degree 1 in two independent variables x, y . Therefore, by Euler's Theorem,

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = v.$$

$$\text{This } \Rightarrow x \left(\sec^2 u \frac{\partial u}{\partial x} \right) + y \left(\sec^2 u \frac{\partial u}{\partial y} \right) = \tan u$$

$$\left(\because \frac{\partial v}{\partial x} = \sec^2 u \frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} = \sec^2 u \frac{\partial u}{\partial y} \text{ and } v = \tan u, \text{ by equation (1)} \right)$$

$$\begin{aligned}\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{\tan u}{\sec^2 u} \\ &= \frac{\sin u}{\cos u} (\cos^2 u) \\ &= \sin u \cos u \\ &= \frac{1}{2} (2 \sin u \cos u) \\ &= \frac{1}{2} \sin 2u.\end{aligned}$$

Example 4 : If $u = \cos^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$.

Solution : Given that

$$u = \cos^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}.$$

$$\text{This } \Rightarrow \cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}} = \frac{x \left[1 + \frac{y}{x} \right]}{\sqrt{x} \left[1 + \sqrt{\frac{y}{x}} \right]} = x^{\frac{1}{2}} \frac{\left[1 + \frac{y}{x} \right]}{\left[1 + \sqrt{\frac{y}{x}} \right]} = x^{\frac{1}{2}} \phi \left(\frac{y}{x} \right), \text{ where } \phi \text{ is a function of } \frac{y}{x}.$$

$\Rightarrow \cos u$ is a homogeneous function of degree $\frac{1}{2}$ in two independent variables x, y .

Let $v = \cos u$ (1)

Then v is a homogeneous function of degree $\frac{1}{2}$ in two independent variables x, y . Therefore, by

Euler's Theorem,

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = \frac{1}{2} v.$$

$$\text{This } \Rightarrow x \left(-\sin u \frac{\partial u}{\partial x} \right) + y \left(-\sin u \frac{\partial u}{\partial y} \right) = \frac{1}{2} \cos u$$

$$\left(\because \frac{\partial v}{\partial x} = -\sin u \frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} = -\sin u \frac{\partial u}{\partial y} \text{ and } v = \cos u, \text{ by equation (1)} \right)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \frac{\cos u}{\sin u}$$

$$= -\frac{1}{2} \cot u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0.$$

Example 5 : If $u = \sin^{-1} \left(\frac{x}{y} \right) + \tan^{-1} \left(\frac{y}{x} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

Solution : Given that

$$u = \sin^{-1} \left(\frac{x}{y} \right) + \tan^{-1} \left(\frac{y}{x} \right) \quad \text{.....(1)}$$

$$\text{Let } \sin^{-1} \left(\frac{x}{y} \right) = \theta.$$

$$\text{Then } \sin \theta = \frac{x}{y}.$$

$$\text{This } \Rightarrow \tan \theta = \frac{x}{\sqrt{y^2 - x^2}}$$

$$\Rightarrow \theta = \tan^{-1} \frac{x}{\sqrt{y^2 - x^2}}$$

$$\Rightarrow \sin^{-1}\left(\frac{x}{y}\right) = \tan^{-1} \frac{x}{\sqrt{y^2 - x^2}} \quad \dots(2)$$

Substituting the value of $\sin^{-1}\left(\frac{x}{y}\right)$ from equation (2) in equation (1), we get

$$u = \tan^{-1} \frac{x}{\sqrt{y^2 - x^2}} + \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1} \frac{\frac{x}{\sqrt{y^2 - x^2}} + \frac{y}{x}}{1 - \left(\frac{x}{\sqrt{y^2 - x^2}}\right)\left(\frac{y}{x}\right)}$$

$$= \tan^{-1} \frac{x^2 + y\sqrt{y^2 - x^2}}{x\sqrt{y^2 - x^2} - xy}$$

$$\text{This } \Rightarrow \tan u = \frac{x^2 + y\sqrt{y^2 - x^2}}{x\sqrt{y^2 - x^2} - xy} = \frac{x^2 \left[1 + \frac{y}{x} \sqrt{\left(\frac{y}{x}\right)^2 - 1}\right]}{x^2 \left[\sqrt{\left(\frac{y}{x}\right)^2 - 1} - \frac{y}{x}\right]} = \frac{\left[1 + \frac{y}{x} \sqrt{\left(\frac{y}{x}\right)^2 - 1}\right]}{\left[\sqrt{\left(\frac{y}{x}\right)^2 - 1} - \frac{y}{x}\right]} = x^0 \phi\left(\frac{y}{x}\right),$$

where ϕ is a function of $\frac{y}{x}$.

$\Rightarrow \tan u$ is a homogeneous function of degree zero in two independent variables x, y .

Let $v = \tan u \quad \dots(3)$

Then v is a homogeneous function of degree zero in two independent variables x, y . Therefore, by Euler's Theorem,

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 0.$$

$$\text{This } \Rightarrow x \left(\sec^2 u \frac{\partial u}{\partial x} \right) + y \left(\sec^2 u \frac{\partial u}{\partial y} \right) = 0$$

$$\left(\because \frac{\partial v}{\partial x} = \sec^2 u \frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} = \sec^2 u \frac{\partial u}{\partial y} \text{ and } v = \tan u, \text{ by equation (3)} \right)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{0}{\sec^2 u}$$

$$= 0.$$

Example 6 : If $u = \sin(\sqrt{x} + \sqrt{y})$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2}(\sqrt{x} + \sqrt{y}) \cos(\sqrt{x} + \sqrt{y})$.

Solution : Given that

$$u = \sin(\sqrt{x} + \sqrt{y}) \quad \dots\dots(1)$$

This $\Rightarrow \sin^{-1} u = \sqrt{x} + \sqrt{y} = \sqrt{x} \left[1 + \sqrt{\frac{y}{x}} \right] = x^{\frac{1}{2}} \phi\left(\frac{y}{x}\right)$, where ϕ is a function of $\frac{y}{x}$.

$\Rightarrow \sin^{-1} u$ is a homogeneous function of degree $\frac{1}{2}$ in two independent variables x, y .

Let $v = \sin^{-1} u \quad \dots\dots(2)$

Then v is a homogeneous function of degree $\frac{1}{2}$ in two independent variables x, y . Therefore, by

Euler's Theorem,

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = \frac{1}{2} v.$$

This $\Rightarrow x \left(\frac{1}{\sqrt{1-u^2}} \frac{\partial u}{\partial x} \right) + y \left(\frac{1}{\sqrt{1-u^2}} \frac{\partial u}{\partial y} \right) = \frac{1}{2} \sin^{-1} u$

$$\left(\because \frac{\partial v}{\partial x} = \frac{1}{\sqrt{1-u^2}} \frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} = \frac{1}{\sqrt{1-u^2}} \frac{\partial u}{\partial y} \text{ and } v = \sin^{-1} u, \text{ by equation (2)} \right)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} (\sin^{-1} u) (\sqrt{1-u^2})$$

$$= \frac{1}{2} (\sqrt{x} + \sqrt{y}) \left(\sqrt{1 - \sin^2(\sqrt{x} + \sqrt{y})} \right) \quad (\text{using equation (1)})$$

$$= \frac{1}{2} (\sqrt{x} + \sqrt{y}) \cos(\sqrt{x} + \sqrt{y}).$$

Exercises

1. Verify Euler's Theorem when $u = x^3 \log \frac{y}{x}$.
2. If $u = \sin \sqrt{\frac{x-y}{x+y}}$, prove that Euler's formula $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$ holds good.
3. If $u = \sin^{-1} \left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right)$, prove that $\frac{\partial u}{\partial x} = -\frac{y}{x} \frac{\partial u}{\partial y}$.
4. If $\sin u = \frac{x^3 + y^3}{x + y}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \tan u$.